



Ancestral Skies Collaboration

Communication 6

Thank you for your help in optimising our observations for the Near Earth Objects. We would now like your advice on managing the technical staff who keep our telescopes operating.

AU1 is part of a cluster of telescopes located on traditional lands where there is evidence that the indigenous inhabitants have lived for tens of thousands of years. The Ancestral Skies Collaboration has negotiated an Indigenous Land Use Agreement to build and operate the telescopes on country, providing a range of financial and non-financial benefits to the indigenous community in exchange for their consent to establish the telescope on their traditional lands. This agreement provides sustainable and intergenerational benefits in areas including training and education. At least 70% of the technical staff for the telescopes will come from the local community, with appropriate skills training provided.

Each week we have a range of tasks that need to be completed by the technical staff, as given in this data file: [Ancestral.py](#)

We have identified the most important skill required by a technician to complete each task, and the number of staff required. For example, the first task is "Electronics diagnostics and repair" and this requires strength in electronics maintenance and a team of two staff to complete.

We have evaluated the strengths of our 13 staff in each of the skills required for our tasks, with higher scores indicating stronger skills, and we add these together for any allocation of staff to a team for a particular task. For example, Technician 0 has a score of 19.0 for the skill of electronics maintenance, while Technician 1 has a score of 11.4. If we allocated Technician 0 and Technician 1 to that first task of "Electronics diagnostics and repair" then the total skill score for completing that task would be 30.4.

Each task is given an estimated duration in hours. We would like to know how to best make use of our technical staff by allocating them to tasks so that the total skill score for all the tasks during the week is as high as possible, noting that each individual technician cannot work more than 36 hours in the week.

How should we allocate the technicians to the tasks? Please give us the optimal total skill score for all the tasks during the week.

2676.9

Communication 7

Due to kinship avoidance practices, some of our staff cannot work in a team together on the same task. We are aware of the following pairs who are required to avoid each other:

Staff	3 & 8	6 & 7	3 & 12	8 & 10
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Unfortunately it seems your current allocation has some of these pairs allocated to the same task. Could you revise your allocation to take these avoidance practices into account? Please give us the updated optimal total skill score for all the tasks during the week.

2615.4

Communication 8

Our staff can only be asked to work a maximum of 10 hours in a day but we have realised that is difficult to plan in the current allocation. We have now revised our list of tasks to be explicit about on which day that task should be completed, as given in this data file:

[Ancestral2.py](#)

In addition to the limit of 10 hours in a day, each technician must have at least two days free from work each week.

Could you revise your allocation to take this daily schedule into account? Please give us the updated optimal total skill score for all the tasks during the week.

2530.4

Communication 9

It has been brought to our attention that some of the staff are doing the "Nightly telescope operation" task but are then being asked to do other tasks during the day. Could you revise the allocation so that if someone does "Nightly telescope operation" then that is their only task for the day?

Please give us the updated optimal total skill score for all the tasks during the week.

2489.3

Communication 10

As part of our commitment to the community, we currently have the opportunity to provide training programs for five of the staff before the week starts. Each training program will focus on a chosen skill, effectively increasing the score for that skill by 7 points for the recipient. Who should we select for this training and which skills should each of them focus on?

Please give us the updated optimal total skill score for all the tasks during the week.

We look forward to reading your final report.